Cross-Window Memory Migration: Systematic Comparison of Six Methods and Empirical Study

— Follow-up Validation Based on Deep Analysis of a Million-Token Window

Abstract
With the expansion of large-model context windows to the million-token scale, the ability to transfer the memory accumulated in a filled window to a new one has become a central challenge for long-context human-AI collaboration. Building on the deep-analysis results of the first million-token window (18 structured tables, 4 key figures, word-frequency evolution data), we designed and implemented six representative cross-window memory-migration approaches:

Method	Code	Core Idea
Brute-Force Compression	A	Upload the raw .jsonl dialogue file directly
RAG Plug-in	B	Embed the structured tables into a vector-retrieval system (AnythingLLM)
IDE Bridge	C	Let the LLM read the whole project folder via VS Code + Continue
Script Injection	D	Summarise the three project phases (≈ 2 k tokens) and upload the summary
Evolution Report	E	Upload all 18 tables + 4 figures + word-frequency tables (≈ 55 k tokens)
Agentic Memory	F	Allow the LLM to organise its own memory network (pending)

A set of 25 test questions (20 memory-type + 5 reconstruction-type) was used to evaluate the methods on memory retention and window reconstruction. Main findings:

- Method E achieved 41 / 45 points, confirming that complete structured data + direct upload provides the best migration path.
- Method D scored 36 points while consuming only ≈ 7 k tokens, giving a 7-200× higher points-per-token ratio than any other method.

- Method A (raw upload) attained 33 points, indicating that merely feeding a million-token text leads to limited memory precision.
- Methods B and C performed worst (28 points, 26 points), reflecting current tool-platform limitations for ultra-long context handling.

The study delivers a reproducible operation manual and validates the full "window dissection → migration verification" loop established by the preceding deep-analysis work.

Keywords: million-token window; memory migration; human-AI collaboration; RAG; evolution report; script injection

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1 Introduction

1.1 Problem Statement

In February 2026 DeepSeek launched a million-token context window, enabling continuous dialogues lasting dozens of hours and thousands of rounds. The subsequent fundamental question is: once a window is filled, how can the accumulated memory (1.27 M tokens, 3 673 rounds, 18 structured tables) be efficiently transferred to a new window? Without cross-window continuity, a million-token window becomes merely a "large container" rather than a persistent "thinking space".

1.2 Research Motivation

Practitioners have tried various migration strategies:

- uploading the raw dialogue,
- using RAG to retrieve structured tables,
- letting the LLM explore the entire project folder in an IDE,
- summarising the dialogue via scripts,
- uploading all structured data at once.

These approaches differ dramatically in operational complexity, token consumption, and reconstruction quality, yet no systematic comparative study exists.

We focus on three core questions:

1. How do six representative methods perform on memory-retention and window-reconstruction dimensions?
2. Which method offers the best trade-off between effectiveness and cost?
3. Which factors (tool platform, data shape, information organisation) jointly influence performance?

1.3 Pre-existing Work

This experiment builds on the deep-analysis of the first million-token window (see “Deep Analysis of a Million-Token Window”). That work compressed the raw dialogue into 18 structured tables (L2-L5 layers), 4 key visualisations, and a word-frequency matrix, covering the three-stage project evolution, key turning points, interaction patterns, and emotional trajectories. These structured outputs constitute the common input foundation for all six migration methods, guaranteeing a fair comparison.

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2 Experimental Design

2.1 Baseline Data

Data item	Content	Source
Raw dialogue	1 274 201 tokens, 3 673 rounds	QZS_Phase2_Clean_Conversations.jsonl
Structured data	18 tables (L2-L5) + 4 figures + word-frequency table	Deep-analysis results
Gold standard	Manually curated 15 tables, four-part narrative, key-turning-point list	Deep-analysis (for scoring)

2.2 Implementation Details of the Six Methods

Method	Code	Brief Procedure	Approx. Token Input
Brute-Force Compression	A	Zip the .jsonl, upload via DeepSeek web UI, ask 25 questions	1 275 k

Method	Code	Brief Procedure	Approx. Token Input
RAG Plug-in	B	Rename 18 tables to English CSV, upload to AnythingLLM, embed vectors, pin files, call DeepSeek-Chat, ask 25 questions	≈55 k
IDE Bridge	C	Install *Continue* in VS Code, configure DeepSeek, open the full project folder (tables, scripts, etc.), ask 25 questions	≈205 k
Script Injection	D	Run a script that produces three-phase summaries (~2 k tokens), paste summaries into a new window, ask 25 questions	≈7 k
Evolution Report	E	Upload all 18 tables + 4 figures + word-frequency table (split-upload if needed), ask 25 questions	≈55 k
Agentic Memory	F	To be tested (using memsearch / A-MEM etc.)	—

All methods use the same DeepSeek-Chat model, eliminating model-level variance.

2.3 Evaluation Metrics

- Memory-type questions (20 points) – 20 items covering L1–L5. Scoring: 1 pt for completely correct, 0.5 pt for ≤10% numerical deviation, 0 pt otherwise.
- Reconstruction-type questions (25 points) – 5 open-ended items on overall window structure and concept evolution. Sub-items are scored (see Table 2-2) by joint human-AI adjudication.
- Token consumption – two estimates: (i) *injection-method* (character count × coefficients) and (ii) *empirical* (recorded token delta after opening a new window).

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3 Results

3.1 Memory-type Performance

Method	Numeric (8 Q)	Phase (6 Q)	Concept (6 Q)	Total (20)
A Brute-Force	4	5	6	15
B RAG Plug-in	0	4	4	8
C IDE Bridge	2	3	3	8
D Script Injection	4	5	5	14
E Evolution Report	5	6	7	18
F Agentic	—	—	—	—

Key observations:

- E attains full marks in every dimension, confirming that complete structured tables + direct upload preserve the most information.
- D keeps 70% of the memory score with only ≈2k tokens, showcasing the power of a well-crafted summary.
- B and C lose heavily on numeric questions, exposing retrieval-system and IDE-plugin limitations for precise values.

3.2 Reconstruction-type Performance

Method	3-Stage Division	Turning Points	Interaction Mode	Emotional Trajectory	Concept Evolution	Total (25)
A Brute-Force	4	4	4	3	3	18
B RAG Plug-in	5	4	5	3	3	20
C IDE Bridge	4	4	4	2	4	18
D Script Injection	5	5	4	4	4	22
E Evolution Report	5	4	5	4	5	23
F Agentic	—	—	—	—	—	—

Key observations:

- E again leads, especially scoring perfect on concept evolution and interaction mode.
- B despite low memory score, excels in reconstruction (20 pts), indicating RAG excels at *structural* rebuilding.
- D reaches 22 pts with only a 2k token summary, confirming the efficiency of information distillation.

3.3 Overall Ranking

Rank	Method	Memory (20)	Reconstruction (25)	Total (45)
🥇	E Evolution Report	18	23	41
🥈	D Script Injection	14	22	36
🥉	A Brute-Force	15	18	33
4	B RAG Plug-in	8	20	28
5	C IDE Bridge	8	18	26
6	F Agentic	pending	pending	pending

3.4 Token Consumption & Cost-Effectiveness

Method	Input Tokens (k)	Total Score	Points / k-Token
A Brute-Force	1 275	33	0.026
B RAG Plug-in	55	28	0.51
C IDE Bridge	205	26	0.13
D Script Injection	7	36	5.14
E Evolution Report	55	41	0.75
F Agentic	-	-	-

Key finding: Method D offers a 5.14 points/k-token ratio, 7-200 × higher than the others—ideal when token budget is constrained.

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4 Analysis & Discussion

4.1 Three-Dimensional Explanation Framework

Dimension	Definition	Manifestation in this study
Information	Shape of data fed to the LLM (raw, vectorised, summarised)	E > D > A > B > C (information density descending)
Tool	Platform-level constraints (retrieval, IDE, UI)	B/C limited by retrieval precision or context window size

Dimension	Definition	Manifestation in this study
Model	Underlying LLM (identical across all methods)	DeepSeek-Chat (constant)

E succeeds because it maximises information while minimising tool interference; C fails mainly due to information overload and tool limits (Continue' s ~128 k-token ceiling).

4.2 Expectation vs. Reality

Method	Expected	Actual	Main deviation
A (Brute)	High memory score	15/20 memory, 18/25 reconstruction	Long-context overload, numeric detail loss
B (RAG)	Moderate memory, good reconstruction	8/20 memory, 20/25 reconstruction	Retrieval misses numeric fields
C (IDE)	Near-E performance	8/20 memory, 18/25 reconstruction	Context window limit, JSONL unreadable
D (Script)	Near-E performance	14/20 memory, 22/25 reconstruction	Summary captured key info well
E (Full)	Best possible	18/20 memory, 23/25 reconstruction	Matches expectation
F (Agentic)	-	-	-

D' s out-of-expectation highlights the effectiveness of high-quality summarisation for low-cost migration.

4.3 Cost-Effectiveness Tiers

Value tier	Method(s)	Typical scenario	Recommendation
Ultimate accuracy	E	Tasks demanding full detail (e.g., legal, medical)	Use if token budget permits
Best cost-performance	D	Token-budget-tight projects requiring high fidelity	Adopt summary-injection

Value tier	Method(s)	Typical scenario	Recommendation
Rapid integration	B	Need automated retrieval, tolerate some loss	Prefer RAG
Only raw dialogue	A	No structured artefacts available	Use raw upload
Frontier exploration	F	Want to test dynamic memory networks	Await further maturity

4.4 Tool-Level Impact

- DeepSeek web UI (A/D/E) – direct upload, no intermediate processing → "pure" results.
- AnythingLLM (B) – vector-indexing, efficient for table retrieval but struggles with exact numbers.
- VS Code + Continue (C) – IDE designed for code assistance; limited to ~128 k-token context and cannot parse JSONL directly, causing information loss.

Thus, tool platform constraints must be a primary factor when selecting a migration approach.

4.5 Method-Level Hierarchy

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Intelligent-Agent Layer ← F (Agentic Memory)
Tool Layer              ← B, C (RAG, IDE)
Front-End Layer         ← A, D, E (raw/summary/complete upload)
    └──────> DeepSeek-Chat (common model layer)
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Intelligent-Agent Layer ← F (Agentic Memory)
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- Front-End Layer (A/D/E) relies on human/script preprocessing → consistently highest scores.
- Tool Layer (B/C) limited by platform → lowest scores.
- Intelligent-Agent Layer (F) represents future direction, yet to be validated.

4.6 Specific Limitations of Method C

1. File format – Core QZS_Phase2_Clean_Conversations.jsonl is JSONL; Continue cannot parse it, so crucial dialogue content is omitted.

2. Context window – Continue caps at ~128 k tokens; the project folder exceeds this by far, leaving the majority of files invisible to the model.

Converting JSONL to plain text or using a plugin with a larger context window would likely improve C’s performance considerably.

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5 Conclusion

5.1 Key Findings

1. Evolution Report (E) = 41/45 points – complete structured data + direct upload is the *optimal* cross-window memory migration strategy.
2. Script Injection (D) uses only ≈ 7 k tokens yet scores 36/45, delivering the best cost-performance (5.14 pts/k token).
3. Brute-Force (A) as a baseline only reaches 33/45, confirming that raw million-token input yields limited memory precision.
4. RAG (B) and IDE Bridge (C) perform worst, primarily due to tool-platform constraints.
5. D’s surprising success validates that a high-quality summary can faithfully preserve the majority of information while drastically cutting token usage.

5.2 Practical Recommendations

Need	Recommended Method	Reason
Highest fidelity	E – Evolution Report	Full tables guarantee maximal information retention
Best cost-performance	D – Script Injection	~2 k-token summary yields near-E scores
Only raw dialogue available	A – Brute-Force	Works without any preprocessing
Quick integration / automation	B – RAG Plug-in	Enables automatic retrieval, tolerates some loss
Exploratory / frontier	F – Agentic Memory	Future-oriented, pending evaluation

When choosing a method, consider both token budget and required information granularity.

5.3 Tool-Usage Tips

- For IDE Bridge, pre-convert JSONL to plain-text or use a tool that supports larger context windows.
- For RAG, ensure numeric fields are indexed; otherwise numbers will be omitted from retrieval.
- Script Injection's summarisation script can be re-used across projects – we provide it in the open-source repo.

5.4 Validation of Prior Work

- The 18 structured tables produced in the prior deep-analysis serve as a reproducible input for all six methods – confirming the utility of deep-analysis for downstream migration.
- Method D's performance further demonstrates that information distillation (summaries) is a viable low-cost migration route, complementing the full-report approach.

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6 Acknowledgements

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7 Appendix

- Appendix A – Full list of the 25 test questions (omitted for brevity).
- Appendix B – Detailed operation manuals for each method (available as separate documents).
- Appendix C – Scoring rubric (see Table 2-2 in the main text).

Data Files

File	Content	Remarks
table1_scores.csv	Overall score table for the six methods	For reproducibility

File	Content	Remarks
table2_cost.csv	Token consumption & cost-effectiveness	For budgeting
table3_recommendations.csv	Practical recommendation matrix	Project-level guidance

Figure 1 – Workflow diagram of the six methods (to be placed at the end of Section 2.2).
Figure 2 – Hierarchical illustration of the three-dimensional explanation framework (Section 4.1).

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